A_5 character table

$\#G = 60$

Conjugacy classes	$\{e\}$	$\{(1\,2\,3)\}$	$\{(1\,2)(3\,4)\}$	$\{(1\,2\,3\,4\,5)\}$	$\{(1\,5\,4\,3\,2)\}$
size	1	20	15	12	12
triv	1	1	1	1	1
std					

$H < G$ $\qquad$ V G-irrep. with $V \subset \mathrm{Ind}_H^G \rho$ for some H-irrep ρ

G finite gp. $k = \bar{k}$ char 0. $\qquad$ V G-irrep.

We know $V \cong_G V^\vee \iff \chi_V = \chi_{V^\vee} (= \bar{\chi}_V$, i.e. χ_V real-valued$)$

So $\mathrm{Hom}_G(V^\vee, V) = 1$

$= \mathrm{Hom}_k(V^\vee, V)^G$

$= (V \otimes V)^G$

$= (\mathrm{Sym}^2 V \oplus \Lambda^2 V)^G$

1-d subspace of $\mathrm{Sym}^2 V \oplus \Lambda^2 V$, so must lie entirely in one of the summands. We say V is
- orthogonally self-dual if $\subset \mathrm{Sym}^2 V$
- symplectically self-dual if $\subset \Lambda^2 V$.

One way to check: if $\psi : V^\vee \xrightarrow{\sim} V$ then

$\psi^\vee : V^{\vee\vee} \xrightarrow{\sim} V$.

V is
- OSD if $\exists \psi$ st. $\psi = \psi^\vee$
- SSD if " " $\psi = -\psi^\vee$.

Proof. $\mathrm{Hom}_k(V^\vee, V) \xrightarrow{\sim} V \otimes V$

$\psi \longmapsto \sum_i \psi(f_i) \otimes e_i$, where $\{e_i\}$ basis V, $\{f_i\}$ dual basis.

$\psi^\vee \longmapsto \sum_i e_i \otimes \psi(f_i)$

if $\psi = \psi^\vee$ then $\psi \in \mathrm{Sym}^2 V$

if $\psi = -\psi^\vee$ then $\psi \in \Lambda^2 V$. $\square$

Made with Goodnotes

recall $\chi_{\Lambda^2 V}(g) = \frac{1}{2}\left(\chi_V(g)^2 - \chi_V(g^2)\right)$. $\chi_{\mathrm{Sym}^2 V}(g) = \frac{1}{2}\left(\chi_V(g)^2 + \chi_V(g^2)\right)$.

Prop. V irrep is SSD if $\langle \mathrm{triv}, \chi_{\Lambda^2 V}\rangle \neq 0$

Proof.
$$\langle \mathrm{triv}, \chi_{\Lambda^2 V}\rangle = \frac{1}{|G|}\sum_{g\in G} \frac{1}{2}\left(\chi_V(g)^2 - \chi_V(g^2)\right) \neq 0$$
$$\| $$
$$\dim_k \mathrm{Hom}_G(\mathrm{triv}, \Lambda^2 V)$$
$$\| $$
$$\dim_k \mathrm{Hom}_k(k, \Lambda^2 V)^G$$
$$\| $$
$$\dim_k (\Lambda^2 V)^G. \qquad \square$$

similar for OSD:

Prop OSD if $\langle \mathrm{triv}, \chi_{\mathrm{Sym}^2 V}\rangle \neq 0$.

Thm (Maschke) G finite. k char 0. every G-rep is semisimple.

Proof. Let V be a G-rep and $W \subset V$ G-subrep. we show W has G-stable complement. Let $\tilde{p}: V \to V$ be a projector onto W; that is
$$\tilde{p}^2 = \tilde{p} \quad \text{and} \quad \mathrm{im}(\tilde{p}) = W.$$

Let $p: V \to V$
$$v \longmapsto \frac{1}{|G|}\sum_{g\in G} g\,\tilde{p}(g^{-1}\cdot v).$$

Claim: p is G-equivariant proj onto W.

1. $$p(h\cdot v) = \frac{1}{|G|}\sum_{g\in G} g\cdot\tilde{p}(g^{-1}h\cdot v)$$
$$= \frac{1}{|G|}\sum_{h^{-1}g\in G} h\cdot h^{-1}g\cdot\tilde{p}((h^{-1}g)^{-1}\cdot v)$$
$$= h\cdot p(v)$$

2. $w\in W$. $$p(w) = \frac{1}{|G|}\sum_{g\in G} g\cdot\tilde{p}(\underbrace{g^{-1}\cdot w}_{\in W}),$$ so $\tilde{p}(g^{-1}w) = g^{-1}\cdot w$
so $\frac{1}{|G|}\sum_{g\in G} g\cdot\tilde{p}(g^{-1}\cdot w) = w$.

hence $p|_W = \mathrm{id}_W$

3. $\mathrm{im}(p)\subset W$: $\tilde{p}(g^{-1}\cdot w)\in W$ so $p(w)\in W$.

4. $p^2 = p|_W\circ p = p$.

So p is projector onto W. Claim: $\ker(p)\subset V$ is G-complement of W.
Clearly $\ker p \cap W = 0$. For any $v\in V$, write $v = \underbrace{p(v)}_{\in W} + \underbrace{(v-p(v))}_{\in \ker p}$ as $p^2 = p$. $\square$

$\ker p$ G-stable as $g\cdot p = p\cdot g$.

Made with Goodnotes

$$Cl(G) = \{ f : G \to k : f(g) = f(h^{-1} g h) \}.$$

$$\psi, \varphi \in Cl(G) \rightsquigarrow \langle \psi, \varphi \rangle = \frac{1}{|G|} \sum_{g \in G} \psi(g) \overline{\varphi(g)}.$$

Thm. V, W G-reps. $\langle \chi_V, \chi_W \rangle = \dim_k \mathrm{Hom}_G(V, W)$.

Proof.
$$\begin{aligned}
\dim_k \mathrm{Hom}_G(V,W) &= \dim_k \mathrm{Hom}_k(V,W)^G \\
&= \mathrm{tr}\left(\tfrac{1}{|G|}\textstyle\sum e_g : \mathrm{Hom}_k(V,W) \to \mathrm{Hom}_k(V,W)\right) \\
&= \frac{1}{|G|} \sum_{g \in G} \mathrm{tr}\left(e_g : \mathrm{Hom}_k(V,W) \to \mathrm{Hom}_k(V,W)\right) \\
&= \frac{1}{|G|} \sum_{g \in G} \chi_{V^\vee \otimes W}(g) \\
&= \frac{1}{|G|} \sum_{g \in G} \overline{\chi_V(g)} \cdot \chi_W(g) \\
&= \langle \chi_W, \chi_V \rangle \quad \in \mathbb{Z} \\
&= \langle \chi_V, \chi_W \rangle. \quad \square
\end{aligned}$$

as $\frac{1}{|G|} \sum e_g$ is a projector onto $\mathrm{Hom}_k(V,W)^G$.

Cor. V, W G-irreps, $k = \bar{k}$. Then $\langle \chi_V, \chi_W \rangle = \begin{cases} 1 & V \cong W \\ 0 & \text{else} \end{cases}$.

Proof. Schur's lem.

Cor. # irreps $\leq$ # conj. classes.

Proof. $\mathbb{1}_g(h) = \begin{cases} 1 & g \text{ conj } h \\ 0 & \text{else} \end{cases}$ is basis for $Cl(G)$.

but irreps are linearly indepent. $\square$

ex. S_3

class	$\{e\}$	(12)	(123)
size	1	3	2
triv	1	1	1
sign	1	-1	1
std	2	0	-1

Thm. $\exists$ iso of G-reps
$$\begin{aligned} ev : \bigoplus_{\rho_i \text{ irreps}} \rho_i \otimes \mathrm{Hom}_G(\rho_i, V) &\xrightarrow{\sim} V. \\ v \otimes f &\longmapsto f(v). \end{aligned}$$

Proof. write $V \cong \bigoplus_{\rho_j \text{ irreps}} \rho_j^{\oplus n_j}$ by Maschke. To show ev surj, enough to show

$$ev : \rho_j \otimes \mathrm{Hom}_G(\rho_j, \bigoplus \rho_j^{\oplus n_j}) \to \rho_j^{\oplus n_j} \quad \forall j.$$

given $(v_1 \cdots v_{n_j}) \in \rho_j^{\oplus n_j}$, define
$$\begin{aligned} f_i : \rho_j &\longrightarrow \rho_j^{\oplus n_j} \quad (1 \leq i \leq n_j.) \\ v &\longmapsto (0 \cdots \underset{i}{v} \cdots 0) \end{aligned}$$

Then $ev(\sum v_i \otimes f_i)$
$= (v_1 \cdots v_{n_j})$ ✓

now count dim: $\dim_k V = \sum_{\rho_i} n_i \dim \rho_i$.

$\sum_{\rho_i} \dim(\rho_i) \cdot \dim \mathrm{Hom}_G(\rho_i, V)$, and $\dim \mathrm{Hom}_G(\rho_i, V) = \dim \mathrm{Hom}_G(\rho_i, \bigoplus_j \rho_j^{n_j})$
$= \sum_j n_j \dim \mathrm{Hom}_G(\rho_i, \rho_j)$
$= n_i$. $\square$

Made with Goodnotes

Cor. $V \cong \bigoplus_{\rho} \rho^{\langle \chi_\rho, \chi_V \rangle}$

Proof. $\rho \otimes \mathrm{Hom}_G(\rho, V) \cong \rho^{\langle \chi_\rho, \chi_V \rangle}$ ↳ dim. count.

Cor. $V \cong W \iff \chi_V = \chi_W$

Proof. immediate

Cor. $\langle \chi_V, \chi_V \rangle = \sum_{\rho_i} n_i^2$, $V = \bigoplus_{\rho_i} \rho_i^{\oplus n_i}$.

Proof.
$$\langle \chi_V, \chi_V \rangle = \langle \sum_i n_i \chi_{\rho_i}, \sum_j n_j \chi_{\rho_j} \rangle$$
$$= \sum_{i,j} n_i n_j \langle \chi_{\rho_i}, \chi_{\rho_j} \rangle$$
$$= \sum_i n_i^2. \quad \square$$

Cor. V irrep $\iff \langle \chi_V, \chi_V \rangle = 1$.

Proof.
$$\sum_i n_i^2 = 1 \iff n_i = 1 \text{ for exactly } 1 \text{ } i. \quad \square$$

Thm. $\gamma : Cl(G) \longrightarrow Z(k[G])$ if ρ irrep, then $\gamma_\psi : \rho \to \rho$ is multip.
$$\psi \longmapsto \sum_{g \in G} \psi(g) \cdot e_g. \quad \text{by } \frac{|G|}{\dim \rho} \langle \psi, \chi_{\rho^\vee} \rangle.$$

Proof. By Schur, γ_ψ is scalar λ and
$$\lambda \cdot \dim \rho = \mathrm{tr}(\gamma_\psi).$$
$$= \mathrm{tr}\left(\sum_{g \in G} \psi(g) e_g\right)$$
$$= \sum_{g \in G} \psi(g) \, \mathrm{tr}(e_g : \rho \to \rho)$$
$$= \sum_g \psi(g) \chi_\rho(g)$$
$$= |G| \langle \psi, \chi_{\rho^\vee} \rangle. \quad \square$$

Cor. $(\chi_\rho)_{\rho \text{ irrep}}$ is ONB for $Cl(G)$.

Proof. we know they are lin ind. remains: span.

Let $\psi \in \mathrm{span}(\chi_\rho)^\perp$. Then $\gamma_\psi : \rho \to \rho = 0$ $\forall \rho$ irrep.
$$\Rightarrow \gamma_\psi : k[G] \to k[G] = 0.$$
$$\gamma_\psi(e) = \gamma_\psi = 0. \quad \text{so } \psi = 0. \quad \square$$

Cor. # irreps = # conj classes

Cor. $\frac{\dim \rho_i}{|G|} \gamma_{\chi_{\rho_i^\vee}}$ acts on ρ_j by $\begin{cases} id & \text{if } i=j \\ 0 & \text{else.} \end{cases}$

Proof. we know it is mult. by $\frac{\dim \rho_i}{|G|} \cdot \frac{|G|}{\dim \rho_j} \langle \chi_{\rho_i^\vee}, \chi_{\rho_j^\vee} \rangle = 0$ if $\rho_i \not\cong \rho_j$
1 if $i=j$.

Made with Goodnotes

$H \leq G$. $\rho : H \to GL(V)$.

$\mathrm{Ind}_H^G \rho := k[G] \otimes_{k[H]} V$, where $k[G] \in \text{mod-}k[H]$ by

$e_g \cdot e_h = e_{gh}$.

- basis $(e_g \otimes v_i)_{g \in G,\ i=1 \cdots n}$.

equiv: $\{ \phi \in \mathrm{Hom}_k(k[G], V) : \phi(gh^{-1}) = h\phi(g) \text{ for } g \in G, h \in H \}$.

Indeed $k[G] \in \text{mod-}k[H] \rightsquigarrow k[G]^* \in k[H]\text{-mod}$

$$
\begin{aligned}
e_h \cdot f_g(e_{g'}) &= f_g(e_{g'} \cdot e_h) \\
&= f_g(e_{g'h}) \\
&= \delta_{g, g'h} \\
&= \delta_{gh^{-1}, g'} \\
&= f_{gh^{-1}}(e_{g'})
\end{aligned}
$$

So $e_h \cdot f_g = f_{gh^{-1}}$.

$k[G] = \bigoplus_{g \in G/H} e_g k[H]$.

so $\mathrm{Ind}_H^G \rho = \bigoplus_{g \in G/H} e_g V$, where $g \cdot (e_{g_i} v)$

$= g_{j(i)} h_i v$

Left:

$$\mathrm{Hom}_{k[G]}(k[G] \otimes_{k[H]} V, W) \cong \mathrm{Hom}_{k[H]}(V, W|_H)$$

indeed

$$
\begin{aligned}
&\overset{*}{=} \mathrm{Hom}_{k[H]}(V, \mathrm{Hom}_{k[H]}(k[G], W)) \\
&= \mathrm{Hom}_H(V, W|_H).
\end{aligned}
$$

(*) Tensor Hom: $R \to S$ i.e. $S \in R$-alg.

$M \in S\text{-mod-}R$.

$$\mathrm{Hom}_S(M \otimes_R V, W) \cong \mathrm{Hom}_R(V, \mathrm{Hom}_S(M, W)).$$

$$f \longmapsto (v \mapsto f(m \otimes v))$$

$$(m \otimes v \mapsto g(v)(m)) \longleftarrow g$$

Made with Goodnotes

Right: $\mathrm{Hom}_H(V|_H, W) \cong \mathrm{Hom}_G(V, \mathrm{Hom}_H(k[G], W))$.

$e_h \cdot e_g = e_{gh^{-1}}$.

Indeed $f: V \to W \longmapsto (v \longmapsto (g \longmapsto f(g^{-1}v)))$

s.t. $h \cdot f(v) = f(h \cdot v)$

$(v \longmapsto F(v)(e)) \longleftarrow\!\shortmid F$

Thm (Artin) $\bigoplus_{H<G \text{ cyclic}} Cl(H) \twoheadrightarrow Cl(G)$.

$\chi_\rho \longmapsto \chi_{\mathrm{Ind}_H^G \rho}$

Proof. we show adjoint $Cl(G) \longrightarrow \bigoplus_{H<G} Cl(H)$ injective.

$\chi_\rho \longmapsto \chi_{\mathrm{Res}_H^G \rho}$

if $\chi_{\mathrm{Res}\,\rho} = 0$ $\forall H < G$ cyclic, then $\forall$

$g \in G$, $\langle g \rangle$ cyclic and $\chi_\rho(g) = \chi_{\mathrm{Res}^G_{\langle g \rangle}\rho}(g) = 0$.

Thm (Artin-Wedderburn) R semisimple k-alg. Then $R \cong \bigoplus \mathrm{Mat}_{n_i \times n_i}(D_i)$ div. alg.

Proof. $R \in R$-mod is semisimple, i.e.

$R \cong \bigoplus M_i^{\oplus n_i}$ simple R-mod.

$R \cong \mathrm{Hom}_R(R,R) \cong \mathrm{Hom}_R(\bigoplus M_i^{n_i}, \bigoplus M_i^{n_i})$

$\cong \bigoplus_i \mathrm{Hom}(M_i, M_i)^{\oplus n_i^2}$ Schur.

$\cong \bigoplus \mathrm{Mat}_{n_i \times n_i}(\underbrace{\mathrm{Hom}_R(M_i, M_i)}_{\text{div alg by Schur.}})$ $\square$

rmk. if $k = \bar{k}$, $D_i = k$. (Schur).

Thm. $k = \bar{k}$. $k[G] \cong \bigoplus_{\rho_i \text{ irrep}} \rho_i \otimes \rho_i^\vee$ as $G \times G$ reps + as rings

where $G \times G \circlearrowright k[G]$ $(g_1, g_2) e_g = e_{g_1 g g_2^{-1}}$

$\circlearrowright \rho_i \otimes \rho_i^\vee$ $(g_1, g_2)(v \otimes w) = g_1 v \otimes g_2 w$.

Proof. we know $k(G) \cong \bigoplus_{\rho_i} \rho_i \otimes \mathrm{Hom}_G(\rho_i, k[G])$.

Claim: $\mathrm{Hom}_G(\rho_i, k[G]) \xrightarrow{\sim} \rho_i^\vee$

$f \longmapsto$ (coeff. of e_e in $f(v)$).

Made with Goodnotes

$$\psi : \mathrm{Hom}_G(\rho, k[G]) \xrightarrow{\sim} \rho^\vee$$
$$f \longmapsto ev_e \circ f$$
$$\hookrightarrow ev_e : k[G] \to k$$
$$\sum a_g e_g \mapsto a_e .$$

$$(g \cdot \psi(f))(v) = ev_e \circ f(g^{-1} \cdot v) = \text{coeff of } e \text{ in } f(g^{-1} \cdot v) = g^{-1} \cdot f(v)$$
$$= \text{coeff of } g \text{ in } f(v)$$

$$\psi(g \cdot f)(v) = \psi(v \mapsto f(v) \cdot g^{-1}) = \text{coeff of } e \text{ in } f(v) \cdot g^{-1}$$
$$= \text{coeff of } g \text{ in } f(v).$$

G-hom ✓

bij: $\dim_k \mathrm{Hom}_G(\rho, k[G]) = \dim_k \mathrm{Hom}_G(\rho, \bigoplus_i \rho_i^{\dim \rho_i})$
$$= \dim \rho_i$$
$$= \dim \rho_i^\vee$$

So ets surj: ets nonzero, by Schur.

$f : \rho \to k[G]$ nonzero. Then $\exists v \in \rho$, s.t.

$\rho(v) \neq 0$, i.e. $f(v) = \sum a_g e_g$ some $a_g \neq 0$.

Then $f(g^{-1} v) = g^{-1} f(v) = \sum a_g' e_g$, $a_e \neq 0$.

So $f \longmapsto (v \mapsto a_e) \neq 0$.

ring iso: basis $v_i \rightsquigarrow$ dual basis v_i^*.

$\sum_g v_i^*(g \cdot v_j) e_g \in k[G]$ is basis for $\rho_i \otimes \rho_j^\vee$.

Prop. (1) χ_ρ, ρ G-rep has integral values

(2) if $\psi \in Cl(G)$ has integral vals, then $\gamma_\psi \in Z(k(G))$ integral.

Proof. (1) $\chi_\rho(g)$ is sum of roots of unity.

(2) note $\gamma_{\mathbb{1}_C}$ integral. Indeed, $\gamma_{\mathbb{1}_C} \in Z(\mathbb{Z}[G]) \subset Z(k[G])$ subring

f.g. as gp

So $\langle \gamma_{\mathbb{1}_C} \rangle \subset Z(\mathbb{Z}[G]) \Rightarrow \gamma_{\mathbb{1}_C}$ integral.

Now ψ w/ int. values is $\sum a_C \mathbb{1}_C$ w/ a_C integral. □

Made with Goodnotes

Cor. ρ irrep. $\mu \in Cl(G)$ integral-valued. Then $\frac{1}{\dim\rho}\sum_g \mu(g)\chi_\rho(g)$ integral.

$\overset{!!}{\mu_\rho}$

Proof. we know $\sum_g \mu(g)\, e_g$ integral.

enough to find $Z(k[G]) \longrightarrow k$

sending $\gamma_\mu \longmapsto \mu_\rho$

claim: $Z(k[G]) \longrightarrow \mathrm{End}_G \rho = k$ works.

$$\psi_\mu \longmapsto ? \xrightarrow{\text{this is}} \frac{\mathrm{tr}(\psi_\mu : \rho \to \rho)}{\dim\rho} = \frac{\mathrm{tr}\left(\sum \mu(g)\, e_g : \rho \to \rho\right)}{\dim\rho}$$

$$= \frac{1}{\dim\rho}\sum \mu(g)\, \mathrm{tr}(e_g : \rho \to \rho)$$

$$= \mu_\rho. \quad \square$$

Thm. $\dim\rho \mid |G|$

Proof. $\mu = \chi_{\rho^\vee}$. By cor,

$$\mu_\rho = \frac{1}{\dim\rho}\sum_g \chi_{\rho^\vee}(g)\,\chi_\rho(g)$$

$$= \frac{|G|}{\dim\rho}\langle \chi_\rho, \chi_\rho\rangle$$

$$= \frac{|G|}{\dim\rho} \quad \text{integral. but } \in \mathbb{Q}, \text{ so } \in \mathbb{Z}.$$

Prop. $|G| = p^r$, k char p.

$\Rightarrow G \to GL_n k$ unipotent

Proof. $\forall g \in G$, $(e_g - e_e)^{p^r} = e_g^{p^r} - e_e^{p^r} \in k[G]$

$$= e_{g^{p^r}} - e_e$$

$$= 0.$$

So $e_g - e_e$ nilpotent in $k[G] \Rightarrow e_g - I$ nilpotent matrix

$\Rightarrow e_g$ unipotent matrix. $\square$

Cor. $|G| = p^r$, k char p. every semisimple rep is trivial

Proof.

HW1 <u>Q3.</u>

D_4 — conj classes: 8 elts

	$\{e\}$	$\{r^2\}$	$\{r, r^3\}$	$\{s, sr^2\}$	$\{sr, sr^3\}$
size	1	1	2	2	2
triv	1	1	1	1	1
orientation	1	1	1	-1	-1
	1	1	-1	-1	1
	1	1	-1	1	-1
square.	2	-2	0	0	0

Made with Goodnotes

k infinite. $V \otimes \bar{k} \cong W \otimes \bar{k}$ as G-reps $\Rightarrow V \cong W$ as G-reps

Proof. Let $S = \{ X \in M_{n \times n}(k) : X \rho_V(g) = \rho_W(g) X \quad \forall g \in G \}$

Since $V_{\bar{k}} \cong W_{\bar{k}}$, $\exists X_{\bar{k}} \in S \otimes \bar{k}$, $\det(X_{\bar{k}}) \neq 0$.

pick basis $X_1 \dots X_r$ for S/k (also basis for $S \otimes_k \bar{k} / \bar{k}$).

Let $p : k^r \longrightarrow k$

$(c_1 \dots c_r) \longmapsto \det\left(\sum_{i=1}^{r} c_i X_i\right)$.

Writing $X_{\bar{k}} = \sum_{i=1}^{r} \bar{c}_i X_i$ for $\bar{c}_i \in \bar{k}$,

we have $p(\bar{c}_1 \dots \bar{c}_r) \neq 0$.

So p is not the zero polynomial, hence $\exists\, c_1 \dots c_r \in k$ st.

$p(c_1 \dots c_r) \neq 0$.

Then $X = \sum c_i X_i \in S$ satisfies $\det(X) \neq 0$, so it is iso$/k$. $\square$

$G = SL_2 \mathbb{C} \xhookrightarrow{\rho} GL_2 \mathbb{C}$. Show $\mathrm{Sym}^n(\mathbb{C}^2)$ is irrep $\forall n \geq 0$.

Proof. x, y basis $\mathbb{C}^2 \rightsquigarrow \mathrm{Sym}^n \mathbb{C}^2$ has basis $x^n, x^{n-1}y, x^{n-2}y^2 \dots y^n$ ($\# = n+1$).

$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2 \mathbb{C} \circlearrowright f(x,y) \in \mathrm{Sym}^n \mathbb{C}^2$

$A \cdot x^i y^{n-i} = (Ax)^i (Ay)^{n-i}$.

if $W \subset \mathrm{Sym}^n \mathbb{C}^2$ is G-subrep, then it contains some $x^i y^{n-i}$.

$A_t = \begin{pmatrix} t & \\ & t^{-1} \end{pmatrix}$. $\quad A_t(x^i y^{n-i}) = t^i x^i \, t^{n-i} y^{n-i} = t^n x^i y^{n-i}$.

??? HW1Q5.

Made with Goodnotes

X integral projective scheme of dim ≥ 1 $/k$.

$\mathcal{L} \in \operatorname{Pic} X$ ample. Then $H^0(X, \mathcal{L}^{-1}) = 0$.

Proof. suppose $\exists\, 0 \neq s \in \Gamma(X, \mathcal{L}^{-1})$.

$$\rightsquigarrow \varphi : \mathcal{O}_X \longrightarrow \mathcal{L}^{-1}$$

$$a \in \mathcal{O}_X(U) \longmapsto a \cdot s|_U \in \mathcal{L}^{-1}(U).$$

since $\mathcal{L}_x \cong \mathcal{O}_{X,x}$ integral domain, $\mathcal{L}$ is torsion-free, so φ inj.

$\otimes \mathcal{L}^n$ exact, so $\rightsquigarrow \mathcal{L}^n \longrightarrow \mathcal{O}_X$

$$t \in \mathcal{L}^n(U) \longmapsto t \otimes s^{\otimes n} \in \mathcal{L}^n \otimes (\mathcal{L}^{-1})^n \cong \mathcal{O}_X. \text{ inj.}$$

$$\rightsquigarrow \text{inj } \Gamma(X, \mathcal{L}^n) \longrightarrow \Gamma(X, \mathcal{O}_X).$$

Now Theorem III 5.2 $\Rightarrow$ $\Gamma(X, \mathcal{O}_X)$ finitely-gen k-v.s.

also integral dom, so finite field ext.

Made with Goodnotes

$f: X \to Y$ finite morphism of proj. sch of same dim $/k$.

(a) $f^! \omega_Y^\circ$ is dualizing for X

(b) X, Y nonsing, $k = \bar{k}$. Then $\exists\, t: f_* \omega_X \longrightarrow \omega_Y$.

(a) $f^! \omega_Y^\circ \in QCoh(\mathcal{O}_X) \longleftrightarrow \mathcal{H}om_{\mathcal{O}_Y}(f_*\mathcal{O}_X, \omega_Y^\circ) \in QCoh(f_*\mathcal{O}_X)$.

X nonsingular proj. variety of dim n $/k = \bar{k}$.

$\cup$

Y nonsing. codim p

$$\begin{tikzcd}
\mathcal{I}_Y/\mathcal{I}_Y^2 \arrow[r] & \Omega_X \otimes_{\mathcal{O}_X} \mathcal{O}_Y \arrow[r] \arrow[d, hook] & \Omega_Y \arrow[r] & 0 \\
& \Omega_X^{n-p} \arrow[r] \arrow[d, hook] & \Omega_Y^{n-p} = \omega_Y & \\
& H^{n-p}(X, \Omega_X^{n-p}) \arrow[r, hook, "f"] & H^{n-p}(Y, \omega_Y) \arrow[r, "t_Y"] & k
\end{tikzcd}$$

Since $H^p(X, \Omega^p) \cong H^{n-p}(X, \Omega_X^{n-p})^\vee$, this gives elt $\eta(Y) \in H^p(X, \Omega_X^p)$.

(a) $P \in X$ closed pt. $t_X(\eta(P)) = 1$

$$\begin{tikzcd}
\Omega_X^\circ \arrow[r] \arrow[d, equal] & \Omega_P^\circ \arrow[d, equal] \\
\mathcal{O}_X \arrow[r] \arrow[d, hook] & k(P) \\
\Gamma(X, \mathcal{O}_X) \arrow[r] & k.
\end{tikzcd}$$

then $t_X(\eta(P)) = ev_P(1_X) = 1 \in k$.

(b) $X = \mathbb{P}^n$, so $H^p(X, \Omega_X^p) = k$. $\eta(Y) = \deg Y \cdot 1$

$$t_X(\xi \cdot \eta(Y)) = t_Y \circ \rho(\xi).$$

$$\eta(Y') = \deg(Y') \cdot 1 \in H^p(H, \Omega_H^p)$$

$$\downarrow i^*$$

$$H^p(X, \Omega_X^p)$$

Made with Goodnotes